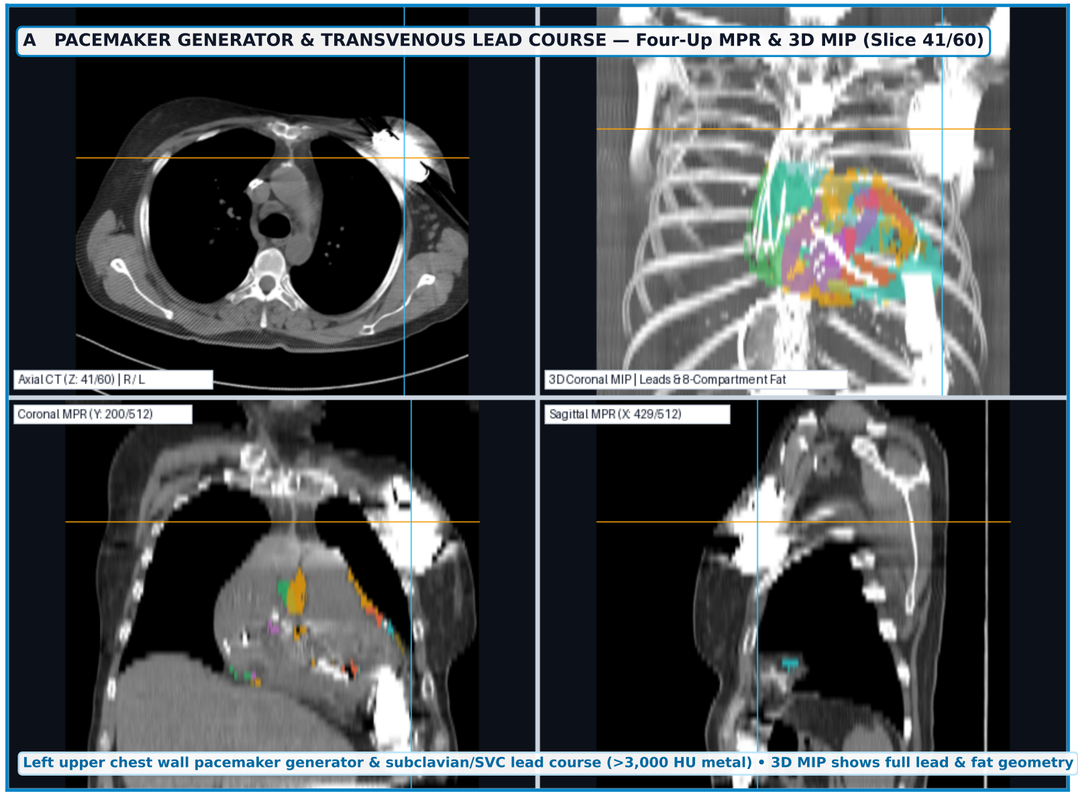
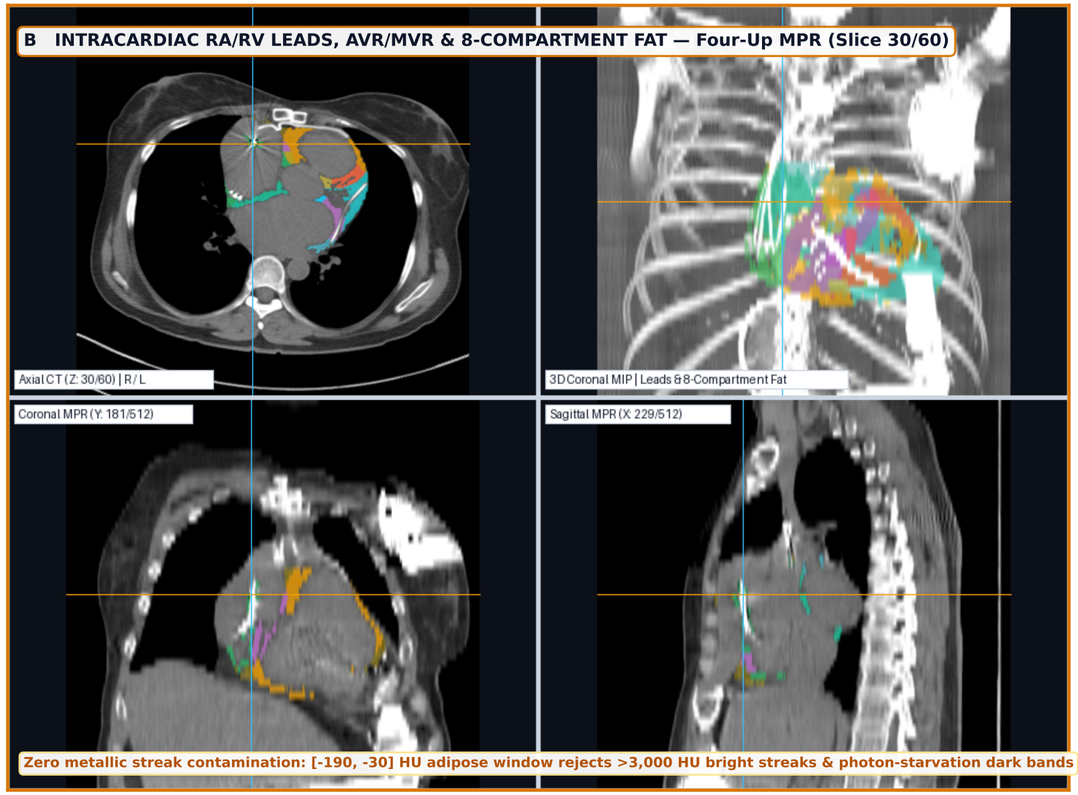
STANFORD COCA CASE STUDY — PARTICIPANT nongated_151: PACEMAKER, VALVE REPLACEMENTS & FAT AI

EAT: 104.2 cm³ • LCX PVAT: -68.0 HU (INFLAMED) • RCA PVAT: -65.6 HU (INFLAMED)

Metallic Streak-Immune 8-Compartment EAT/PVAT Segmentation + NV-Reason-CT Detection of Pacemaker Leads, AVR/MVR & Cardiomegaly

NV-Reason-CT: L. Chest Pacemaker + RA/RV Leads + AVR/MVR + Cardiomegaly





C 8-COMPARTMENT CARDIAC ADIPOSE & CRISP-CT INFLAMMATION QUANTIFICATION

Compartment / Biomarker	Volume (cm³)	Mean HU	Clinical Interpretation / Artifact Status
Total Epicardial Fat (EAT)	**104.2 cm³**	**-64.9 HU**	**Moderate Burden (~54th %ile; Normal < 100 cm³)**
• Right Ventricle (EAT RV)	44.3 cm³	-64.2 HU	RV free-wall fat intact despite RV apex lead
• Left Ventricle (EAT LV)	39.8 cm³	-65.8 HU	LV epicardial pad (Cardiomegaly envelope)
• Right Atrium (EAT RA)	26.1 cm³	-64.1 HU	RA appendage/groove intact despite RA lead
• Left Atrium (EAT LA)	23.2 cm³	-65.5 HU	Periatrial fat adjacent to mitral prosthesis
Total Pericoronary Fat (PVAT)	**23.1 cm³**	**-68.6 HU**	**10mm 3D Anisotropic EDT Perivascular Cuff**
• LAD Pericoronary (PVAT LAD)	6.5 cm³	-74.4 HU	NORMAL (<= -70.1 HU CRISP-CT threshold)
• LCX Pericoronary (PVAT LCX)	**7.8 cm³**	**-68.0 HU**	**ACTIVE INFLAMMATION (> -70.1 HU; +2.1 HU shift)**
• RCA Pericoronary (PVAT RCA)	**9.6 cm³**	**-65.6 HU**	**ACTIVE INFLAMMATION (> -70.1 HU; +4.5 HU shift)**

D NVIDIA NV-REASON-CT (3D ViT + Qwen3.5-4B) — VERBATIM AI RADIOLOGIST REPORT

Medical devices:	**Cardiac pacemaker in left upper chest wall with RA/RV leads. Aortic & mitral valve replacements.**
Lungs & airways:	No focal infiltrates or lung opacities. Trace partial atelectasis in right lung base. Airways patent.
Heart & Pericardium:	**Cardiomegaly. No pericardial effusion. Normal caliber thoracic aorta and main PA.**
IMPRESSION 1:	**1. Cardiomegaly with cardiac pacemaker and aortic and mitral valve replacements.**
IMPRESSION 2-3:	**2. Epicardial adiposity with active coronary microvascular inflammation. 3. Trace RLL atelectasis.**
KEY FINDINGS TAG:	**Medical material, Cardiomegaly, Atelectasis, Coronary artery wall calcification, Postoperative change**

WHY PACEMAKER & PROSTHETIC VALVE RESILIENCE MATTERS IN CARDIAC CT:

1. Metallic Beam-Hardening & Streak Immunity: Intracardiac RA/RV pacemaker leads and prosthetic valves produce >3,000 HU bright streaks and <-300 HU photon-starvation dark bands. Strict [-190, -30 HU] adipose gating rejects both extremes without corrupting LCX/RCA PVAT HU.
2. Comprehensive Post-Operative Context: NV-Reason-CT autonomously identifies the left chest wall pulse generator, dual-chamber RA/RV leads, aortic and mitral valve replacements (AVR/MVR), and cardiomegaly—providing complete clinical context alongside 3D fat biomarkers.